

Hotel Reservation Android App — Project Report

Abstract

This project is a native Android hotel reservation application developed using Java. It provides separate user and administrator workflows covering authentication, room management, booking, payments, and notifications.

User Features

Users can register and log in, recover passwords, access a dashboard, manage or browse rooms, create bookings, complete payment workflows, view payment information, and receive notifications.

Administrator Features

Administrators have a dedicated dashboard with functions for adding and managing rooms, managing bookings, viewing payments, and sending notifications.

Technology Stack

The application uses Java, Android SDK, AndroidX, Material Components, RecyclerView, CardView, Firebase Authentication, Firebase Firestore, Firebase Cloud Messaging, Firebase Functions, Firebase Analytics, Glide, OkHttp, Socket.IO/WebSocket components, PayPal, and Braintree.

Application Structure

The project separates activities, adapters, models, repositories, services, ViewModels, utilities, and WebSocket components, providing a structured Android application architecture.

Backend and Services

Firebase is used for authentication and cloud data services. Firebase Cloud Messaging and WebSocket-related components support notification functionality. Payment integrations include PayPal and Braintree components.

Conclusion

The project demonstrates full-cycle Android application development for hotel reservations, integrating authentication, cloud data, booking management, payment processing, and notifications.

Security

The portfolio copy excludes the original Firebase configuration and project identifiers. Private payment and service credentials should never be committed to a public repository.